

Classical Mechanics 1, Autumn 2025 CMI

Assignment 6

Due by 6pm, 17 Oct, 2025

Damped motion, Spherical polars, Angular momentum

1. **⟨2 + 2 + 1 + 3⟩ Damped motion.** Suppose a particle of mass m moving on the real line with position $x(t)$ is subject to a conservative force arising from the potential $V(x)$ as well as a damping force $-\gamma\dot{x}$ with damping coefficient $\gamma > 0$. (a) Show that the ‘energy’ $E = \frac{1}{2}m\dot{x}^2 + V(x)$ is a nonincreasing function of time. (b) Suppose $V(x) = -\frac{1}{2}kx^2 + \frac{1}{3}gx^3$ with $k = 2, g = 3$. Draw a graph of $V(x)$. Qualitatively describe the nature of the motion for initial conditions (c) $x(0) = \dot{x}(0) = 0$ and (d) $x(0) = 1, \dot{x}(0) = 0$.
2. **⟨5⟩** Consider the integral $\int_0^1 dx/x^a$. Show that the integrand $1/x^a$ has an **integrable divergence** (integrand diverges but integral is finite) at $x = 0$ for $0 < a < 1$ and a nonintegrable divergence for $a \geq 1$. What is the integral when it is finite?
3. **⟨5⟩** Consider a particle moving in 3d Euclidean space. Its position vector is $\mathbf{r}$ relative to a fixed origin. Using a figure of **spherical polar coordinates** (r, θ, ϕ) , express $\hat{r}$ as a linear combination of $\hat{x}, \hat{y}, \hat{z}$. Then express $\hat{r}$ as a linear combination of

$$\begin{aligned}\hat{\theta} &= -\sin\theta\hat{z} + \cos\theta(\cos\phi\hat{x} + \sin\phi\hat{y}) \quad \text{and} \\ \hat{\phi} &= -\sin\phi\hat{x} + \cos\phi\hat{y}.\end{aligned}\tag{1}$$

4. **⟨5⟩** Consider a particle moving in 3d Euclidean space. Its position vector is $\mathbf{r}$ relative to a fixed origin and its momentum is $\mathbf{p} = m\dot{\mathbf{r}}$ where m is its mass. Its **angular momentum** with respect to the chosen origin is defined as $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. The Cartesian components of $\mathbf{L}$ are denoted L_x, L_y, L_z . In spherical polar coordinates, show that the z component of angular momentum L_z of the particle can be expressed as $L_z = mr^2 \sin^2\theta \dot{\phi}$.